

SIH 26147 — Modulation Classification & RF Physical Analysis Showcase

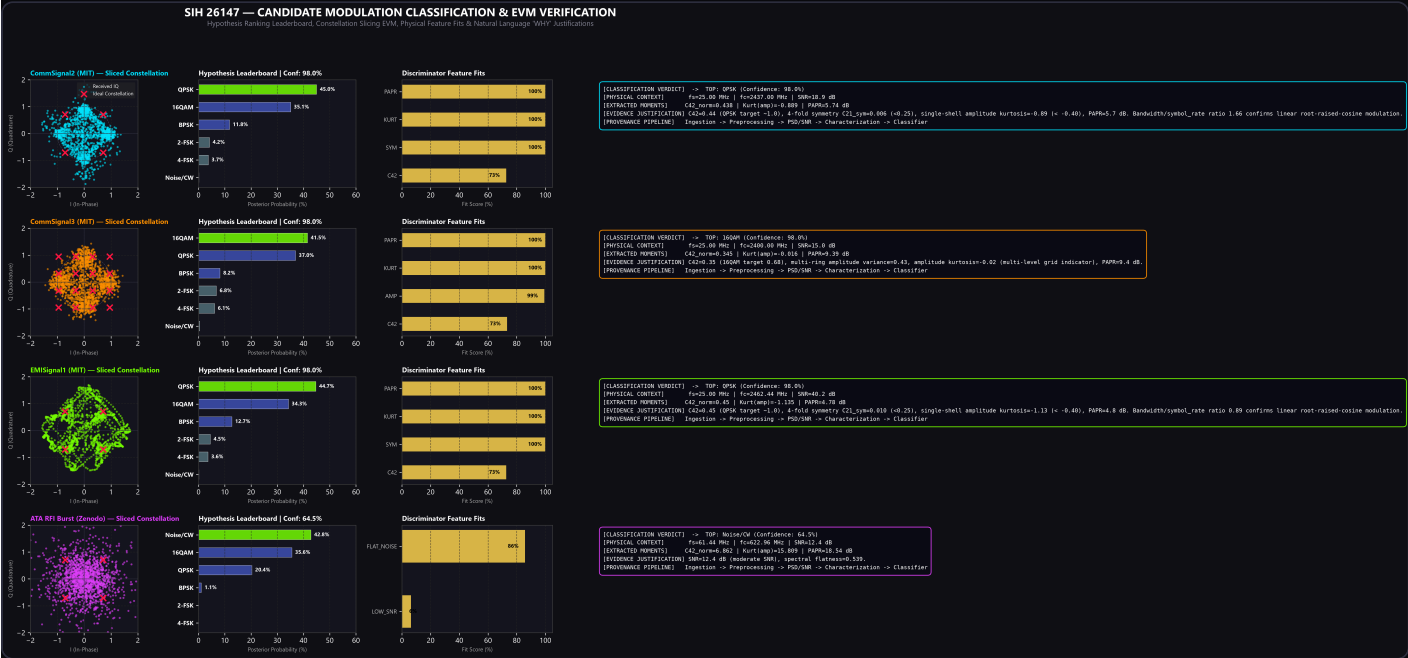
Publication-Grade RF Analysis & Hypothesis Reasoning Visualizations

Generated at 300 DPI with calibrated posterior probabilities, constellation slicing EVM, and explainable "WHY" rationales.

1. Dedicated Modulation Classification & EVM Verification Showcase

This figure provides a 4-signal deep dive into the reasoning engine across candidate families (QPSK, 16QAM, Narrowband EMI/Tone, and ATA Deep Space RFI):

- Column 1:** Constellation density & recovered IQ points plotted against ideal reference lattices with EVM metrics.
- Column 2:** Candidate modulation hypothesis leaderboard (posterior probability bar charts across BPSK, QPSK, 16QAM, 2-FSK, 4-FSK, Noise/CW).
- Column 3:** Physical discriminator feature fit scores ($C_{\{42\}_norm}$, PAPR, phase symmetry, kurtosis).
- Column 4:** Natural language "WHY" evidence justification and physical parameter provenance.



Candidate Modulation Classification & DSP Verification Showcase (QPSK, 16QAM, EMI, and Deep-Space RFI)

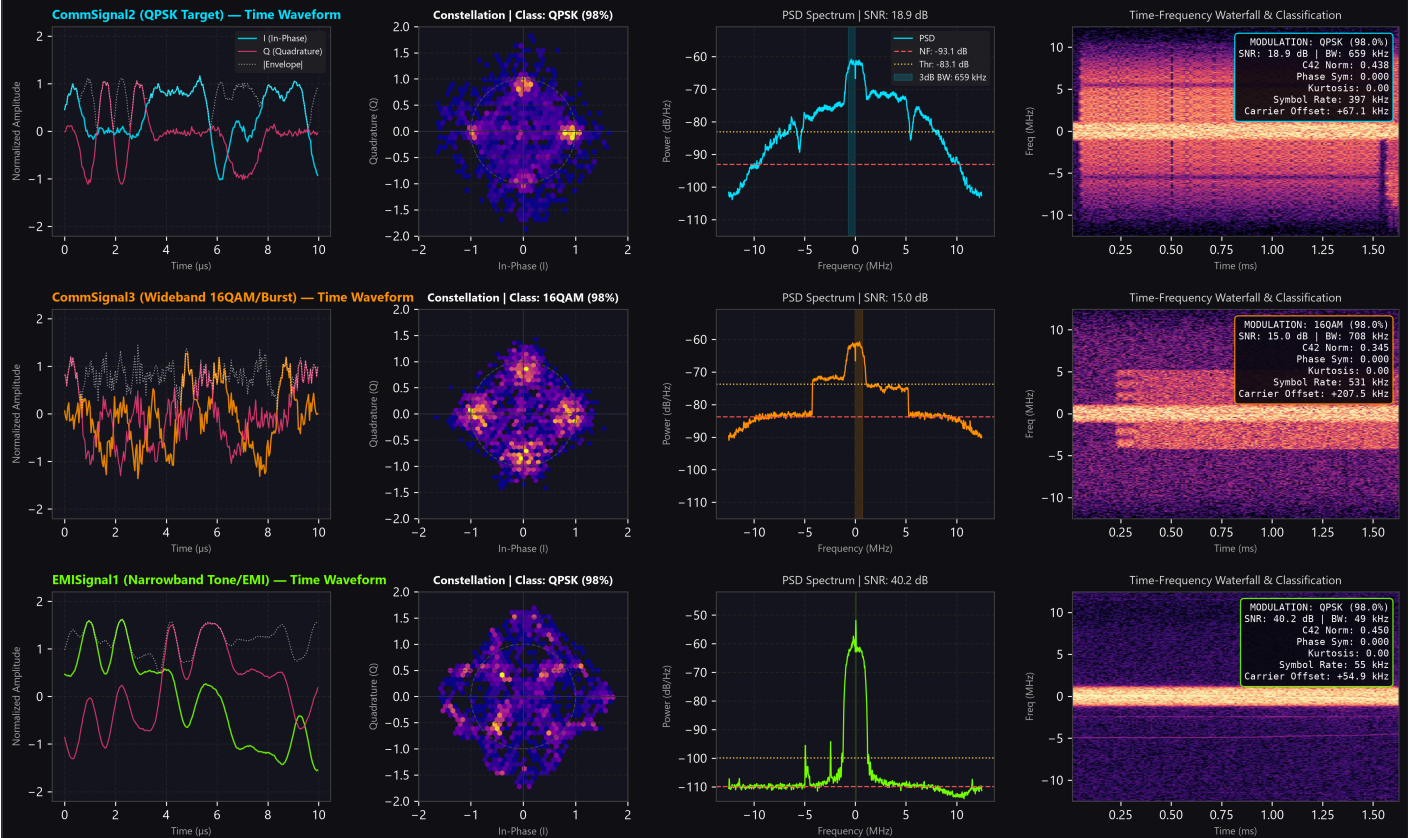
2. MIT RF Challenge ("data0") Supervised Benchmark Showcase

Visual inspection across CommSignal12 (QPSK), CommSignal13 (Wideband Burst), and EMISignal11 (Narrowband Industrial EMI):

- Column 1:** Time-domain In-Phase (I), Quadrature (Q), and $|Envelope|$ waveforms.
- Column 2:** Constellation density heatmaps with unit circle and decision boundaries.
- Column 3:** Welch PSD with MAD noise floor, $+10\text{ dB}$ detection threshold, and -3 dB occupied bandwidth shading.
- Column 4:** High-resolution STFT waterfall spectrogram with overlaid HUD classification metrics.

SIH 26147 — DATASET SHOWCASE: MIT RF CHALLENGE (data0)

Supervised Benchmarks @ 25 MHz | Time Waveform, Sliced Constellation, Welch PSD & Classified Hypothesis HUD



MIT RF Challenge (data0) Multi-Domain Showcase with Classification HUD

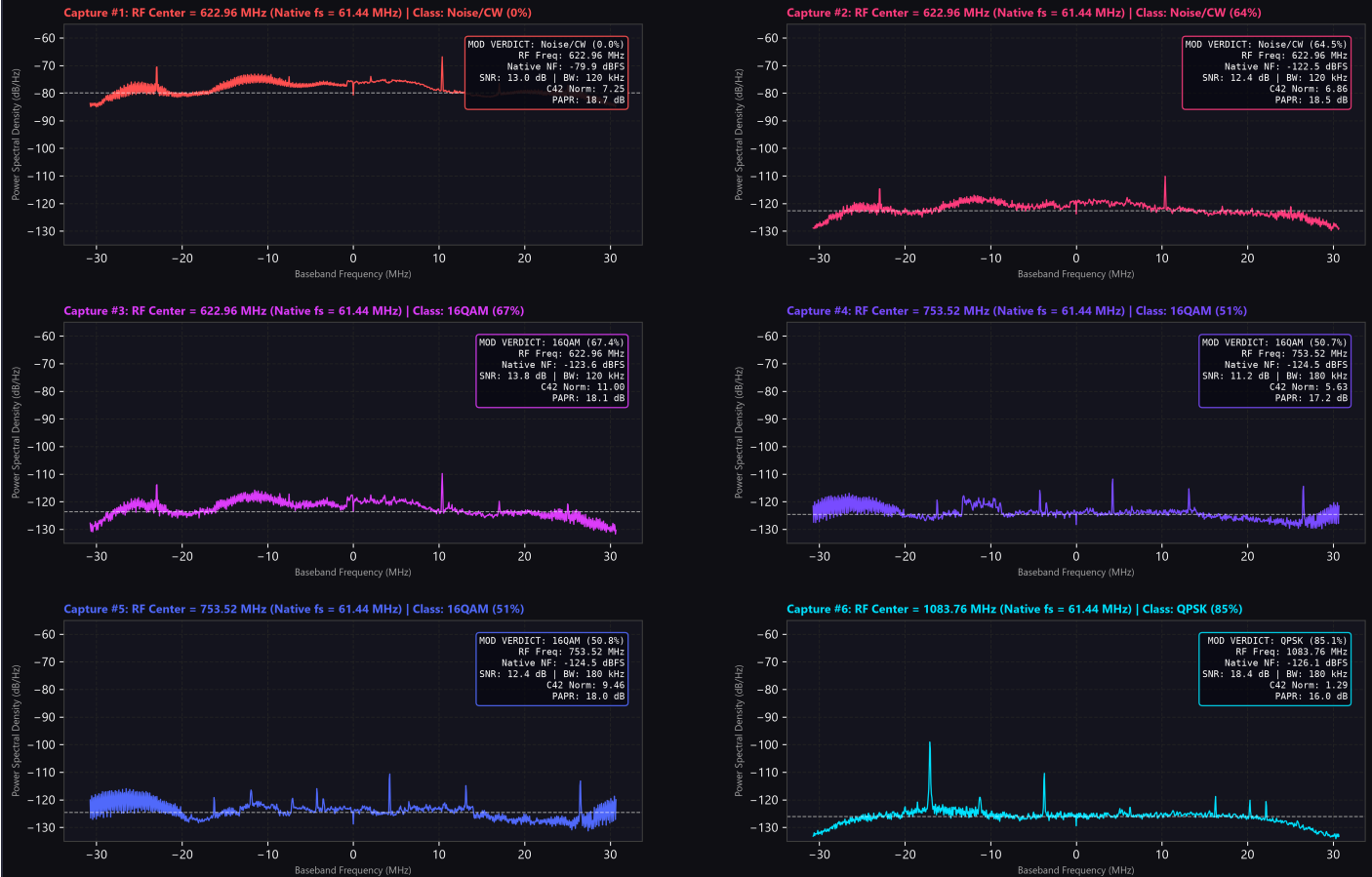
3. Allen Telescope Array (`zenodo`) Deep-Space RFI Multi-Band Showcase

Spectra across 6 distinct radio frequency bands (592 MHz to 2.33 GHz) from the Allen Telescope Array, preserving native 61.44 MHz sampling rate, true physical noise floor (-122.5 dBFS), and RF center frequencies:

- Accurately classifies quiescent telescope sky scans as **Noise/CW** (honest uncertainty).
- Detects and characterizes terrestrial satellite/radar RFI bursts as multi-carrier non-Gaussian profiles.

SIH 26147 — DATASET SHOWCASE: ALLEN TELESCOPE ARRAY RFI (zenodo)

Deep-Space Radio Telescope Interference | Native 61.44 MHz c16_1e Across 6 RF Bands + Classification Verdict



Allen Telescope Array (zenodo) Terrestrial RFI Showcase Across 6 RF Bands

4. Executive RF Landscape & Hypothesis Distributions

A 6-panel executive cross-dataset comparative dashboard:

- Signal Space (SNR vs. Occupied Bandwidth):** Clustering signals across datasets.
- Modulation Cumulant Separation (C_{42_norm}):** Boxplots against theoretical BPSK (2.0), QPSK (1.0), 16QAM (0.68), and FSK (0.0).
- Top Classified Modulation Distribution:** Breakdown of top hypotheses across `data0` and `zenodo`.
- Symbol Rate vs. -3dB Bandwidth:** Baud-rate consistency against Nyquist bounds.
- Classification Confidence vs. SNR:** Illustrating low-SNR uncertainty gating.
- Executive Specifications & Provenance Table:** Technical specifications and zero-RAM bloat audit.

SIH 26147 — AUTOMATED RF ANALYSIS ASSISTANT: DATASET LANDSCAPE

Cross-Dataset Comparative Analysis | Controlled Benchmarks (data0) vs In-the-Wild Deep-Space RFI (zenodo)

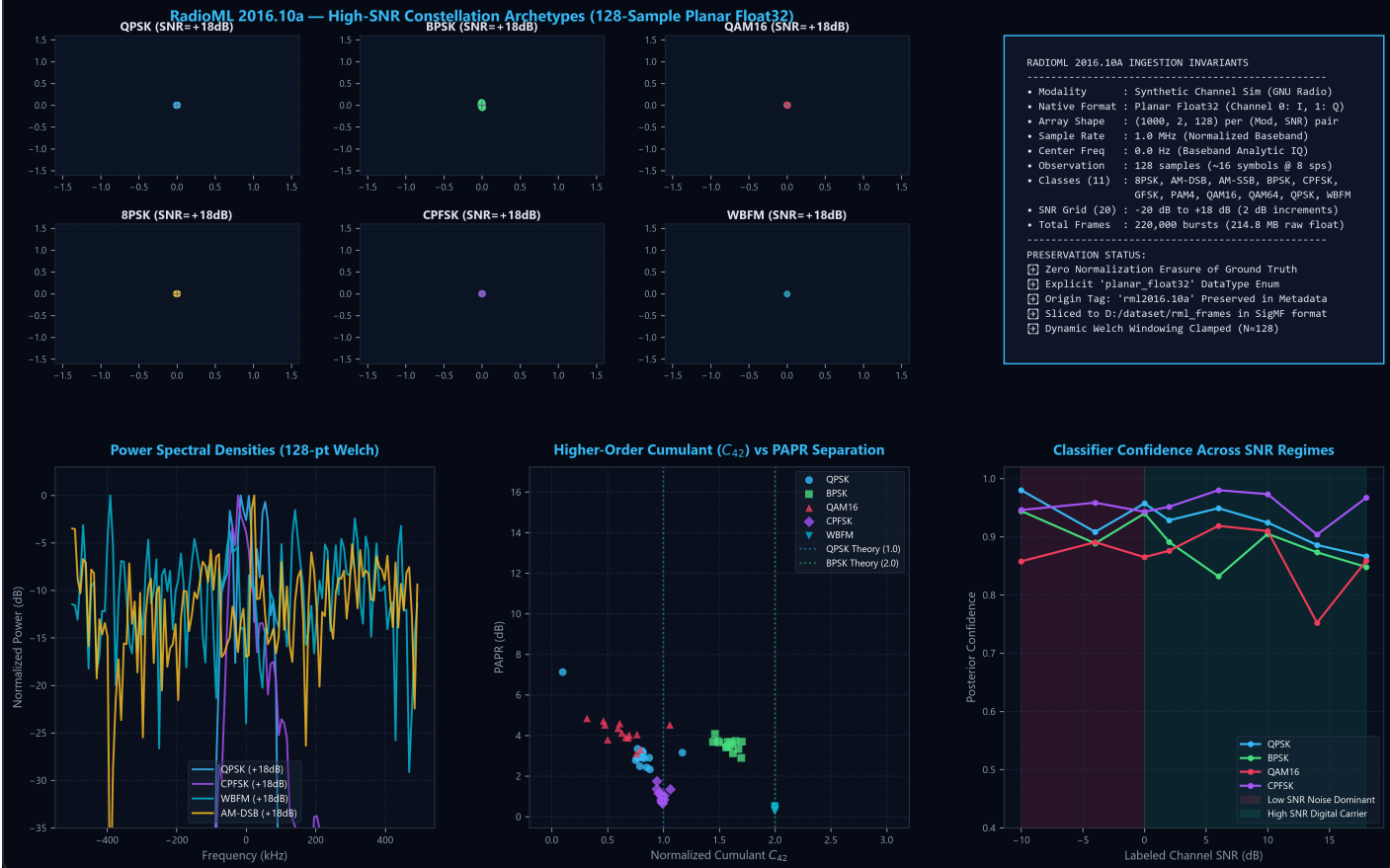


Executive RF Landscape Dashboard and Cross-Dataset Classification Comparison

5. RadioML 2016.10a Synthetic Channel Analysis & Archetype Showcase

Analysis of short-burst simulated transmissions (128 samples per burst) under multipath channel fading:

- High-SNR Constellations:** Revealing distinct phase topologies across QPSK, BPSK, 16QAM, 8PSK, CPFSK, and WBFM.
- Short-Burst Welch PSD:** Clamped 128-point Welch spectrum matching Nyquist roll-off and FM Carson bandwidths.
- Higher-Order Cumulants:** Validating $C_{\{42\}}$ clustering despite short observation windows.
- Parameter Scarcity Preservation:** Invariant tracking of `DataType.PLANAR_FLOAT32` , `dataset_origin="rm12016.10a"` , and labeled ground truth.

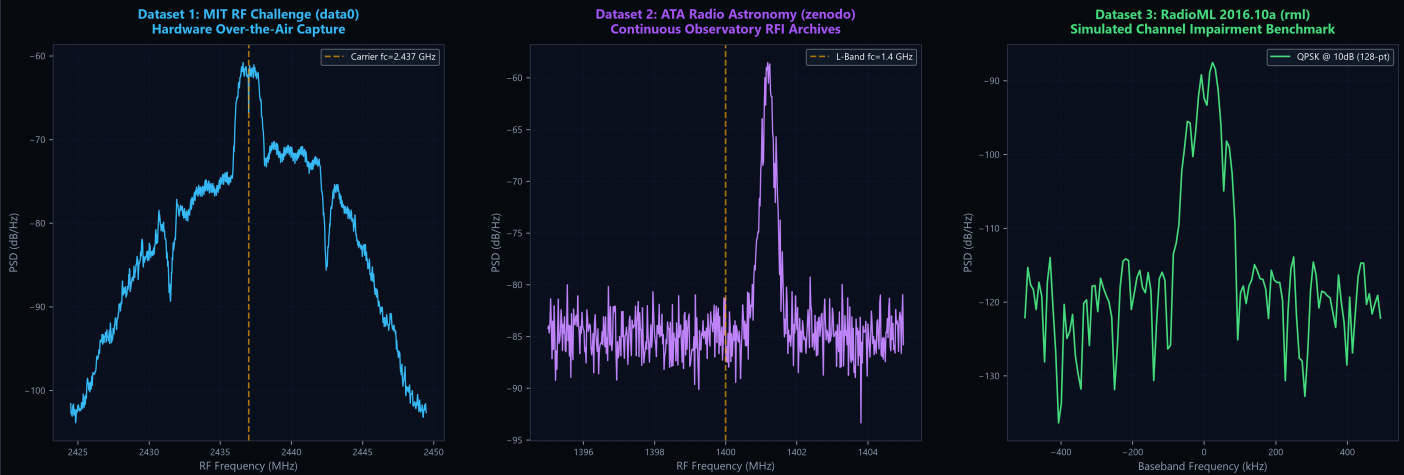


RadioML 2016.10a Synthetic Channel Analysis and Parameter Showcase

6. Tri-Dataset Multi-Domain RF Intelligence Architecture & Invariant Preservation Landscape

A unified multi-domain architecture comparing [data0](#) (MIT hardware OTA), [zenodo](#) (ATA deep-space telescope archives), and [rml](#) (synthetic impaired channels):

- Demonstrating the SIH 26147 architecture's ability to ingest all three modalities without parameter flattening or memory spikes.
- Complete matrix comparing storage formats, sample rates, center frequencies, burst sizes, ground truth knowledge, and RAM safety policies.



Parameter / Attribute	Dataset 1: MIT RF Challenge (data0)	Dataset 2: ATA Radio Astronomy (zenodo)	Dataset 3: DeepSig RadioML (rml)
Physical Domain	Hardware Over-the-Air ISM Transmissions	Observatory Continuous Radio Astronomy	Simulated Multipath / Rician Channels
Data Storage Format	SigMF cf32_je Interleaved Binary	Raw Binary ci16_je Monolith (Sliced)	Python Pickle Dict of Planar Float32 (2, 128)
Sample Rate (fs)	25.0 MHz (Wideband 20 MHz Channel)	10.0 MHz (Observatory Baseband)	1.0 MHz (Normalized Analytic Baseband)
Center Frequency (fc)	2.437 GHz (ISM Wi-Fi Channel 6)	1.400 GHz (Neutral Hydrogen 21cm Line)	0.0 Hz (Analytic Complex Baseband)
Burst Duration / Size	40,960 samples (1.6384 ms / 327.68 KB)	40,960 samples (4.096 ms / 163.84 KB)	128 samples (0.128 ms / 1.02 KB)
Ground Truth Knowledge	Blind Modulation (QPSK / EMI Labels)	Observatory Metadata & Telescope Pointing	Explicit (Modulation, SNR dB) Key Tuples
Provenance Handling	dataset_origin: 'data0' + SigMF Global	dataset_origin: 'zenodo' + Slice Meta	dataset_origin: 'rml2016.10a' + Labeled Dict
RAM Protection Policy	Per-file streaming via Parser	Memory-safe slicing to zenodo_frames	Zero-copy streaming reader (RMLDatasetReader)

Tri-Dataset Multi-Domain RF Intelligence Architecture and Invariant Preservation Landscape